Stephen Purdue

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EDUCATION

Royal Holloway, University of London — **London, UK**
MSc, Artificial Intelligence — *2026 - Expected 2027*

University of Winchester — **Winchester, UK**
BSc, Computer Game Development — *2023 - 2026*

EXPERIENCE

Corvian — **2026 - Present**
Founding Engineer — *London, UK*

- Drove 100% of technical strategy and execution, leading the design and creation of a scalable kitchen management system using Python, FastAPI, and TypeScript.
- Implemented critical systems including containerisation via Docker, AWS Caching and Load Balancing, and Stripe payments.

Mustang Group — **2021 - Present**
Sous-Chef, Chef de Partie — *Southampton, UK*

- Led a kitchen of five through high-volume, deadline-driven service under constant time pressure, requiring rapid problem-solving, attention to detail, and direct communication with other colleagues in an agile environment.
- Mentored and trained 4-8 junior chefs and new starters per intake, maintaining consistent service standards and reducing onboarding time during high staff turnover.

PROJECTS

Stock Market Prediction Engine — **GitHub**

- Developed an LSTM model in PyTorch to forecast daily stock prices from historical time-series analysis data, reducing prediction error by 81% versus a moving-average baseline. Containerised the full stack with Docker Compose, reducing local setup time to under 5 minutes.

UNIX Shell — **GitHub**

- Developed a UNIX style shell in Python, implementing command parsing, built-in commands, environment-variable handling, and input/output redirection. Developed a modular, testable architecture using OS-level process and file-system APIs, stengthening my understanding of Linux systems programming.

Procedural Generation Algorithm — **GitHub**

- Engineered a scalable procedural generation system in Unity using C# and Binary Space Partitioning to recursively subdivide space into room layouts and connective corridors in linear time. Reduced level build-time by 77% by implementing modular, parameter-driven logic.

Adaptive Combat Simulator — **GitHub**

- Developed an adaptive combat simulator, achieving a 75% win rate against a baseline agent, with overall reward improving from 32% to 87%, measured over 30 unique evaluation simulations by training a PPO-agent using Python and TensorFlow.

TECHNICAL SKILLS

Languages: Python, C#, C/C++, SQL, Bash, Elixir, TypeScript
Technologies: Linux, GitHub, Docker, AWS, CI/CD
Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

INTERESTS

- Snowboarding, Competitive Gaming